

## **Maternity**

### **\* Nagele's Rule**

- ✓ To determinate day of labor
  - Day +7
  - Month + 9 or - 3
- First day of last period  
 Ex: let's say = oct 7
- Day 7 + 7 = 14
  - Octubre + 9 = July
- Board = Marzo 4 (bc febrero = 28 days)

### **GTPAL**

|          |                                                 |
|----------|-------------------------------------------------|
| <b>G</b> | - # of Gestation (except Mole Diataforme)       |
| <b>T</b> | - Term = 37- 42 weeks                           |
| <b>P</b> | - Preterm = 20 – 37 weeks                       |
| <b>A</b> | - Abortion = Before 20 weeks (Ectopic pregnant) |
| <b>L</b> | - Live = buscar ese primero para ubicarte       |

Nota; Don't forget to count if is in pregnant consultation

- \* Schedule for Antepartum Evaluation
- If no risk factor:
  - Until 28 – 32 weeks = Every 4 weeks
  - Until 32 – 36 weeks = Every 2 weeks
  - After 36 weeks = weekly

### **\* Pregnancy Sign**

| Presumptive (MOM)                                                                                                                                                                        | Probable (RN)                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Positive (Gynecologist or midwife nurse)                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Amenorrhea<br>2. N/V<br>3. ↑ Size and feeling of breast.<br>4. ↑ urinary Frequency<br>5. Pronounced nipples<br>6. Quickening (first perception of baby) of fetal movement 16-20 weeks | 1. Pregnancy test<br>2. Hegar's test<br>Softening of the lower uterine segment. (6weeks)<br>3. Googell's sign<br>Softening of the cervix (2 months).<br>4. Chadwick's sign (toto morado)<br>Violete bluish discoloration of mucus, vagina and vulva (4weeks)<br>5. Ballotement (16 – 20 ) weeks (da pataditas)<br>Rebounding of the fetus against the examiner finger on palpation.<br>6. Braxton hicks' contraction<br>Irregulars, intermittent, painless (entrada de mes) | 1. Fetal heart rate.<br>- Doppler ultrasound (lo primero para detectar the fetal hearth rate) (10-12) weeks or<br>- Fetoscope after 20 weeks<br>2. Active fetal movement<br>Palpated by midwife or gynecologist.<br>3. Ultrasound<br>4. X-ray. |

**\*Fundal height = Supine Position**

### **Complication to the Mom**

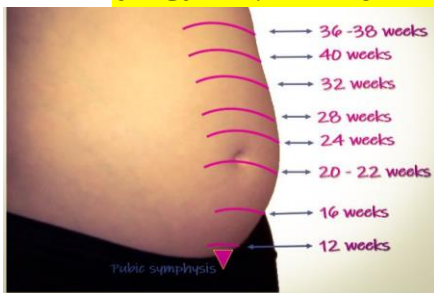
- ✓ Supine hypotension (compresses the inferior vena cava)
- ◇ Place women on side lateral position (choice left)
- ◇ Once it passes go back to the supine position to measure.

#### **\* Steps (SATA)**

1. Place patient in supine position.
2. Place the end of the tape (0) at the level of the symphysis pubis.
3. Stretch the tape to the top of the uterine fundus.
4. Note and record the measurement

#### **- After 20 weeks**

- After 20 weeks 1cm x weeks until 37 weeks
- $\pm 2$  = OK
- Call Dr. If  $+ 2$  or  $- 2$



- SP – 12 weeks
- Between SP and Umbilicus = 16 weeks
- Umbilicus = 20 weeks
- Fundal height more than expected for gestational age could be:
  - Hydatidiform mole (que hace pensar una mole? Que tenga 22 weeks of gestation and 29 fundal high)
  - Twins

### **\* Nutritional Requirements**

- Add 300 Kcal x day during pregnant
- Add 500 Kcal x day during breastfeeding

#### **Diet**

- High calcium = Milk or yogurt 3-4 cup x day
- High Iron = Organ meat, meat with vit C to fix iron
  - ◇ Meats
  - ◇ Shellfish
  - ◇ Orange juice and tomato juice to fix

#### **- Acid Folic replacement:**

- Starting in childbearing age
- To prevent anemia and Neural tube defects
  - 400 mcg x day – childbearing age
  - 600 mcg x day – during pregnant

### **\* Weight during pregnant**

- Normal weight BMI: 18.5 – 25 = 25 – 30 pounds during pregnant
- Under weight BMI: 18.5 = ↑ until 40 pounds

- Overweight BMI > 25 = ↑ until 25 pounds

BMI =  $\frac{\text{Weight Kg}}{(\text{height m})^2}$

Eg 64 kg/ 1.65 m

BMI =  $\frac{64 \text{ kg}}{1.65 \times 1.65}$

BMI = 23.7 = Normal weight

|                                                                                                                                                                                                                                                       |                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>- Underweight = less than 18.5</li> <li>- Normal weight = 18.5 – 25</li> <li>- Overweight = 25 – 29</li> <li>- Obese = &gt;29</li> </ul> <p>Board: who has more risk for cardiopathy?<br/>= Cardiovascular</p> | <p>Ex: Normal weight pregnant = During pregnant ↑25 – 30 pounds</p> <p>140 pounds – 34 weeks – how many pounds?<br/>= 159</p> <p>140 + 5 (before 20weeks)<br/>+ 14 pounds (1xweek after 20 weeks)</p> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

- 1<sup>st</sup> trimester of pregnant = ↑ 3 – 5 pound (in general)

- After 20 weeks = ↑ 1 pound x week until 37 weeks

**Embarazada start with 130 pounds ahora tiene 34 weeks. Cuanto debe de pesar?**

**Assessment (OB) a todas las embarazadas**

#### **TORCH**

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>T</b> | <p><b>Toxoplasmosis</b> - avoid cat feces, avoid contact with little box (board husband change the little box)</p> <ul style="list-style-type: none"> <li>- avoid gardening</li> <li>- Avoid ingestion of unpasteurized milk or undercooked meals</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>O</b> | <p><b>Others : HIV</b></p> <ul style="list-style-type: none"> <li>- ZIDOVUDINE = antiretroviral</li> <li>- Beginning after 14 weeks (P.O)</li> <li>- IV during labor</li> <li>- Syrup newborn for 6 weeks after birth</li> <li>- No internal fetal monitoring if Mom is HIV positive.</li> <li>- When baby born – 1<sup>st</sup> action <ul style="list-style-type: none"> <li>▪ Use gloves before contact with baby</li> <li>▪ Bath newborn</li> <li>▪ Do physical examination</li> <li>▪ Breastfeeding is contraindicated (same in galactosemia)</li> <li>▪ Vaginal delivery if viral load less than 1000</li> </ul> </li> </ul> <p><b>Diagnostic</b></p> <ul style="list-style-type: none"> <li>- Confirmatory diagnostic: 18 – 24 months <ul style="list-style-type: none"> <li>✓ ELISA</li> <li>✓ WESTERN BLOOD</li> </ul> </li> </ul> <p><b>Early Diagnostic</b></p> <ul style="list-style-type: none"> <li>✓ Polymiraze chain reaction test = Positive 3 – 4 months after birth.</li> </ul> <p><b>Hep B- AGHBs</b> (before 20 weeks) if positive:</p> <ul style="list-style-type: none"> <li>- Administer Hep B vaccine (mandatory for all babies + 5 pounds weight) to the newborn if weight more than 5 pounds, and Hep B immunoglobulin in two different site and syringe, in one vastus lateralis the Hep b and the immunoglobulin in the other vastus lateralis at the same time.</li> </ul> <p>- If less than 5 pounds only administer Immunoglobulin.</p> |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>R</b> | <b>Rubella</b> (German measles) (Droplets) <ul style="list-style-type: none"> <li>– Do rubella titer before 28 weeks of gestation. <ul style="list-style-type: none"> <li>○ Less than 1:8 must: (the baby can have malformations) <ul style="list-style-type: none"> <li>▪ Vaccination (mom) after birth within 72 hours of birth</li> <li>▪ Wait 3 months for the next pregnancy</li> </ul> </li> <li>○ If Rubella titer &gt; 1:10 = OK <ul style="list-style-type: none"> <li>▪ No vaccination required bc mom has antibodies.</li> </ul> </li> </ul> </li> </ul> |
| <b>C</b> | <b>Cytomegalovirus</b> - if positive = high risk for malformation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>H</b> | <b>Herpes Simple</b> – (type 2 = genital) can have a vaginal delivered if no active or out break                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

## **Diagnostic Studies**

### **1. Urinalysis (porque le da mucho kidney infection)**

- ✓ Glucose
  - 1+ = Normal
  - More than 2+ = Gestational Diabetes
- ✓ Proteins
  - 1+ = Normal
  - > 2 = pre- eclampsia - Eclampsia
- ✓ Albumins
  - 1+ = Normal
  - > 2 = pre- eclampsia – Eclampsia
- ✓ Leukocyte Esterase or nitrites positive (bacterial infection)
  - UTI
  - Infection (kidneys)

**Note: lymphocytes = viral infections)**

- ✓ Ketonuria
  - Hyperemesis gravidarum (Metabolic Acidosis)
  - Dehydration severe

### **2. Ultrasound**

- ✓ Abdominal
  - With full bladder
- ✓ Transvaginal
  - With Empty bladder
  - To check cervix

### **3. Biophysical profile**

**Noninvasive (20 weeks)**

**To check for Fetal well being**

- a. Fetal Breathing movement. = 2
- b. Fetal movement = 2
- c. Amniotic fluid Index. = 2
- d. Fetal Heart rate = 2
- e. Fetal tone = 2

Between 8 – 10 = Fetal well being

Less than 6 = Call the Dr. immediately bc no well being

### **4. DNA Genetic Testing**

To detect Genetic Diseases

- a. 7 weeks of gestation = # 1.(to detect genetic D)

- b. Trisomy 21 = Down Syndrome
- c. Trisomy 18 = Edward
- d. Trisomy 13 = Patau Syndrome

5. **Chorionic Villus Sampling** (Invasive)

- Non-Maleficence
- At 10 – 13 weeks of gestation to determine for Genetic Disease.

Nota: After Any Invasive procedure if Mother = RH negative administer RHOGAN (doesn't matter dad)

6. **Alpha Fetoprotein.** (blood example)

- 15 – 17 weeks
  - a. If Elevated = Neural tube deficit or spinal bifida
  - b. If decreased = Down syndrome
  - c. If later than 17 weeks = could be false result

7. **Amniocentesis.** (Invasive)

- ✓ Confirmatory for neural tube defect
- ✓ Non-maleficence principle
  - Before 20 weeks = To confirm genetic Abnormalities
    - Full bladder
    - Down Syndrome
    - Neural tube deficit
  - After 25 – 30 weeks = To check for Lung maturity
    - **L/E = 2:1 = OK = Lung Maturity**
    - **With empty bladder**
      - **L/E = less than 2:1 = No lung maturity**
        - Administer STEROID to Mom
          - DEXAMETASONE
          - BECLOMETHASONE

8. **Streptococcal Exam**

- ✓ Vaginal and rectal sample
- ✓ 36 – 37 weeks of gestation

Board: When practice streptococcal exam? = 36 -37 weeks (before due day)

| 9. FERM TEST (especulo)                                                                                                                                                                                                                                                 | 10. NITRAZINE TEST (afuerita)                                                                                                                                                                                        | 11. FIBRONECTIN TEST (cultivo)                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>- Microscopic seidel test to determine the presence of any amniotic fluid.</li> <li>- Sterile technique</li> <li>- Women in lithotomy position</li> <li>- Instruct to cough during exam (to see the fluid coming out)</li> </ul> | <ul style="list-style-type: none"> <li>- Test strip used to detect the presence of amniotic fluid in vaginal secretion</li> <li>- Nitrazine strip or swab turn <b>BLUE</b> = positive for amniotic fluid.</li> </ul> | <ul style="list-style-type: none"> <li>- Cervical and vaginal secretion sampling.</li> <li>- If positive to fibronectin means:</li> <li>- Onset of labor in 1 – 2 weeks after.</li> <li>- <b>16 – 20 weeks = normal expected fibronectin in vaginal secretion</b></li> </ul> |

12. **CBC COUNT, HB, HEMATOCRIT**

- ✓ During 1<sup>st</sup> trimester of pregnant =
  - **Expected CBC 15 000 – 16000**

- During Labor: 20 000 – 25 000 = Expected CBC
  - HB, Hematocrit – 1<sup>st</sup> trimester of pregnancy = 10 – 11
- (board: why in the 1<sup>st</sup> trimester ↓HB? = ↑ circulate volume)
- Physiologic Anemia = Expect – No treatment
    - Diet High in Iron
      - Organ meats (liver, kidney)
      - Green leafy vegetable, shellfish
      - Vit C + orange juice to fix

Nota: if 2<sup>nd</sup> trimester HB = 10. Is low and need treatment

### 13. Blood Type

- If mother RH positive = is ok (no se hace nada)
- If mother RH Negative do exam to the father, and if the father RH negative = is ok (no se hace nada)
- If mother RH negative and father RH positive = Do:
  - Indirect Coombs test to the mother
    - Before 28 weeks
    - If negative = Administer Rogan to the mom at 28 weeks.
  - Do Indirect coombs test after birth to the mom
    - If negative = Administer Rhogam to the mother, Within 72 h of birth
    - If positive = OK

**Administer Rogan always to the mother RH-** (sin importar quien es el padre)

**After:**

- ◇ Abortion
- ◇ Ectopic pregnant
- ◇ Amniocentesis
- ◇ Villius chorionic sampling
- ◇ Hydatidiform mole

### \* Discomfort during pregnant

1. GI, morning sickness (nausea and vomiting in the morning)

#### Intervention

1. Eat dry crackers on arising or a protein bar and stay on bed at least 15 mint after (dry carbohydrate).
2. Ginger Ale
3. Small and frequently feeding
4. No water after meals
5. Antiemetic
  - a. Dimenhydrinate, Dramamine, (grabinol)
  - b. If hyperemesis gravidarum=
    - Emergency = Hospital admission
    - Severe vomiting = dehydration
    - Metabolic Acidosis
2. Backache
  - a. Pelvic tilt or rock exercises
  - b. Abdominal muscles contractions
  - c. Use low heel shoes
  - d. Sleep in firm mattress

- e. Carefully lifting
- 3. Constipation
  - ✓ Treatment = Water
  - = Fiber
  - = Walking
- 4. Leg Cramps = Due to low Ca.
  - ✓ Increase Calcium intake (1 L/ daily)
  - ✓ Feet dorsiflexion
  - ✓ Do not massage
- 5. Pyralism (escupidera)
  - ✓ Suck hard candies
  - ✓ Use lip balm

**\* Bleeding during pregnant**

| Before 20 weeks                                                                                                       | After 20 weeks                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>- Hydatidiform Mole</li> <li>- Ectopic pregnant</li> <li>- Abortion</li> </ul> | <ul style="list-style-type: none"> <li>- Placenta previa</li> <li>- Placental Abrupt</li> </ul> |

**1. Hydatidiform Mole. “No FHR”**

- Is a form of Gestational Trophoblastic disease
- Manifests as an edematous “Grape like Cluster” that may be non-malignant or may develop into choriocarcinoma (endometrial cancer).

Nota: It's not a cancer pero si no se trata can lead to endometrial cancer.

**Assessment**

1. No fetal tone
2. Vaginal bleeding (4-5 week of gestation)
  - ✓ Bright red or dark brown.
3. Fundal height greater than expected for gestational age.
4. ↑↑↑ HCG = chorionic gonadotropin = toomuch elevated
5. Sign of pre-eclampsia = ↑ BP + Proteinuria
6. Hyperemesis gravidarum (severe vomiting) = hospital admission bc severe dehydration.
7. Snowstorm patter on ultrasound

**Intervention**

- a. (D&C) Uterine evacuation (legrado diagnostic)
- b. HCG monitored every 1-2 weeks until normal pregnancy level.
  - ✓ After that every 1-2 months for 1 year
  - ✓ If persistent elevated more than 6 months = Hysterectomy + Chemotherapy (total)
- c. Wait 1 year for next pregnancy

**2. Ectopic pregnancy**

**Cause**

1. Inflammatory pelvic Disease
  - Chlamydia

- Gonorrhea
- 2. Previous tubal surgery
- 3. Congenital anomalies of the fallopian tube.

#### Assessment

1. Amenorrhea (4-8weeks)
2. Unilateral lower quadrant abdominal pain
3. Bleeding (gradual or oozing, no bleeding)

#### Rupture. = **life threatening**

1. Board like abdomen pain radiated to the shoulder
2. Hypovolemic shock
  - ✓ Hypotension + tachycardia

Nota: los dolores que se priorizan en el board son los que se irradian a la izquierda.  
Except this one that if right rupture irradiate in the right shoulder.

#### Intervention

1. IV fluid (NaCl 0.9%) – Priority #1
2. Prepare for surgical intervention
3. Administer **methotrexate** if is stable patient
  - ✓ Avoid sexual intercourse

### 3. Abortion

| Type                                                        | Assessment                                                                                                                                                                      | Nursing Consideration                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Threatened (Amenaza)                                     | <ul style="list-style-type: none"> <li>- Vaginal bleeding</li> <li>- Soft uterus</li> <li>- Cramping</li> <li>- <b>Cervix close</b></li> </ul>                                  | <ul style="list-style-type: none"> <li>- Ultrasound</li> <li>- <b>HCG studies</b></li> <li>- Decreased activities x 24-72 hours.</li> <li>- No sexual intercourse for 2 weeks</li> </ul>                                                                                                 |
| 2. Inevitable or imminent (after pain = se abrio el cuello) | <ul style="list-style-type: none"> <li>- Persistent symptoms</li> <li>- <b>Cervical dilation</b></li> </ul>                                                                     | <ul style="list-style-type: none"> <li>- Complete or incomplete</li> <li>- Decrease HCG</li> </ul>                                                                                                                                                                                       |
| 3. Incomplete                                               | <ul style="list-style-type: none"> <li>- Persistent symptoms</li> <li>- <b>Exposure of part of product of conception</b></li> </ul>                                             | <ul style="list-style-type: none"> <li>- D &amp; C</li> <li>- Administer oxytocin</li> <li>- BT at bedside</li> <li>- Administer medication for bleeding               <ul style="list-style-type: none"> <li>✗ Methylergonovine</li> <li>✗ Hemabate (carboprost)</li> </ul> </li> </ul> |
| 4. Complete                                                 | <ul style="list-style-type: none"> <li>- As above except retained tissue (todo se expulso)</li> <li>- <b>Cervix close or near to close</b></li> </ul>                           | <ul style="list-style-type: none"> <li>- Oxytocin for uterine contraction.</li> <li>- BT set at bed side</li> <li>- Methylergonovine or hemabate for bleeding</li> </ul>                                                                                                                 |
| 5. Missed Abortion (Stillborn)                              | <ul style="list-style-type: none"> <li>- <b>Cervix is closed</b></li> <li>- High risk for infection and DIC = oozing of blood through IV Line = Call Dr. immediately</li> </ul> | <ul style="list-style-type: none"> <li>- Less than 8 weeks = D&amp;C</li> <li>- More than 10-12 weeks = Induction</li> </ul>                                                                                                                                                             |

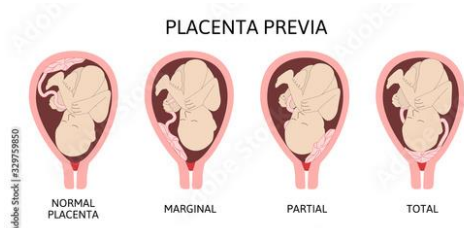


|                                             |  |                                               |
|---------------------------------------------|--|-----------------------------------------------|
| Habitual = (3 or more spontaneous abortion) |  | - Cerclage<br>(encircling cervix with suture) |
|---------------------------------------------|--|-----------------------------------------------|

Nota: Como se que una amenaza de aborto es un aborto inminente = cuando las HCG están decreased

If placenta previa = no sex at all

#### 4. Placenta Previa



- Is an improperly implanted placenta in the lower uterine segment near or over internal cervical os.

#### Types

1. **Total:** The internal cervical os is covered entirely by the placenta. = C section
2. **Partial:** The lower border is within 3 cm of the internal Os but is not fully covered. = C section
3. **Marginal:** The placenta is implanted in the lower uterus (more than 3 cm) but it is not covering the os
  - ✓ This one is the only one that can be a normal birth.
  - ✓ Total and partial must have C- Section.

#### Assessment

1. Sudden onset of **painless, bright red**, vaginal bleeding.
2. Anemia.
3. Hypovolemic shock
  - ✓ BT set at bed site.
4. Uterus soft, relax (non-tender)

#### Intervention

1. Monitor maternal vital sign and FHR
2. Prepare for ultrasound to confirm diagnostic
3. Vaginal examination or any other action that would stimulate uterine activities like sex should be **avoided**.
4. Bed rest = Trendelenburg Left lateral
5. IV fluid
  - ✓ Tocolytic medication in case of risk for preterm labor.
    - ❖ Nifedipine
    - ❖ Terbutaline
    - ❖ Mgso4
    - ❖ Indomethacin
6. Non stress test weekly (contraindicated contraction stress test)
7. Amniocentesis for lung maturity if:
  - ✓ L/E = 1:1 administer steroids
8. Send home:
  - ✓ Limited activity

- ✓ No enemas
- ✓ No douching
- ✓ No sexual intercourse
- 9. Cesarean delivery
  - ✓ Elective

5. **Abrupt Placentae** = (life threatening emergency)

- Premature separation of the placenta from the uterine wall after 20 weeks of gestation (more common during fetus delivery)

**Causes**

- Hypertension
- Cocaine

**Assessment**

1. Sudden onset of dark red vaginal bleeding or not
  - ✓ Absent of visible blood.
2. **Very painful** (tenderness)
3. Uterine tetani (uterine contraction of 90 second or more)
4. Sign of fetal distress
5. Signs of maternal shock

**Intervention**

1. O2, IV fluid, blood products
2. Place in Trendelenburg left lateral
3. Prepare for emergency delivery
- ✓ If minimal or absent variability of FHR less than 5 = C session
4. Check for DIC
  - ✓ Abrupt placentae
  - ✓ Missed abortion

**Gestational Hypertention** (after 20 weeks)

- Detected first time during pregnant = >140/90

| Pre- Eclampsia mild                                                                                                                                          | Pre- Eclampsia moderate to severe                                                                                                            | Warning sign of eclampsia                                                                                                                                                                            | Eclampsia Seizure                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>- BP&gt; 140/90</li> <li>- Proteinuria = 2+</li> <li>- Edema: eyes, face, fingers</li> <li>- Puffy fingers</li> </ul> | <ul style="list-style-type: none"> <li>- BP&gt; 160-110</li> <li>- Proteinuria= 3+ or more</li> <li>- Generalized edema (ascites)</li> </ul> | <ol style="list-style-type: none"> <li>1. Epigastric pain (Priority) <b>RED FLAG</b></li> <li>2. Severe headache</li> <li>3. Blurred vision</li> <li>4. Clonus</li> <li>5. HELLP Syndrome</li> </ol> | <ul style="list-style-type: none"> <li>- Rolling eyes = Convulsion</li> <li>- Tonic clonic movement</li> <li>- convulsion</li> </ul> |

H = Hemoptysis

E = Elevated liver enzymes

L

L

P

} = Low platelets

Emergency Priority (A correr)

### Intervention. (Gestational hypertension)

1. Hospital admission
    - ✓ Bed rest
    - ✓ Decrease environmental stimulation
    - ✓ Restrict visitors
  2. Check for vital sign
    - ✓ RR if less than 12 do not administer MgSO<sub>4</sub>
    - ✓ RR- 16 – 20 = OK
  3. IV fluid
  4. Magnesium Sulfate (MgSO<sub>4</sub>) – therapeutic level = 4- 7 in OB
- Goal:
- to prevent and treat seizure
  - To decreased clonus
- ✓ DTR (2) = Normal (deep tendon reflex)
- After MgSO<sub>4</sub> decrease clonus but continuing with high BP tx: Hydralazine

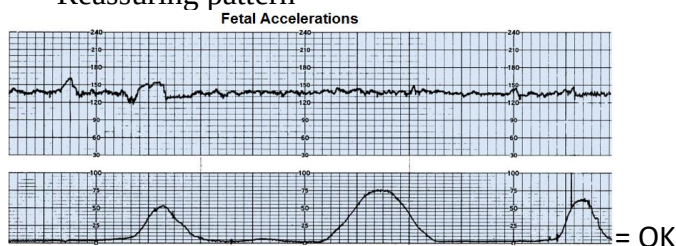
### Magnesium Sulfate intoxication or toxicity = >7

1. Respiratory depression
  - if RR less than 12
  - Stop MgSO<sub>4</sub> + administer calcium gluconate
2. DTR – Decreased = 1 or absent = 0, Normal = 2
  - ✓ +3 = pre- eclampsia – eclampsia, clonus
  - ✓ -1 = MgSO<sub>4</sub> toxicity
3. Decreased urinary output less than 30ml/hr.
  - Have to decreased or stop MagSO<sub>4</sub>
4. CNS Depression (respiratory Acidosis)

| Nonstress test (ultrasound)                                                                                                                                                                                                                                                                                                                                                                                                         | Contraction Stress test or Oxytocin challenge                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>- Use to determine fetal well being</li><li>- Client need to void before.</li><li>- Give hand held pen to the mother</li></ul> <p>And the nurse check the number en el monitor.</p> <ul style="list-style-type: none"><li>- 2/15/15/20 = Reactive = OK</li></ul> <p>= good finding= Acceleration of FHR = ok</p> <p>Non-reactive = Bad = do contraction stress test or Oxytocin challenge</p> | <ul style="list-style-type: none"><li>- IV or nipple stimulation</li></ul> <p>La IV es la mas exacta.</p> <ul style="list-style-type: none"><li>- If positive:<ul style="list-style-type: none"><li>✓ Late deceleration = Utero placentae insufficiency (falta O<sub>2</sub>)</li></ul></li><li>- If Negative = OK = Fetal well being</li></ul> |

### Acceleration of FHR = Good finding

- Reassuring pattern



- Increased FHR due to fetal stimulation 2/15/15/20 = Acceleration = Normal

### Cause

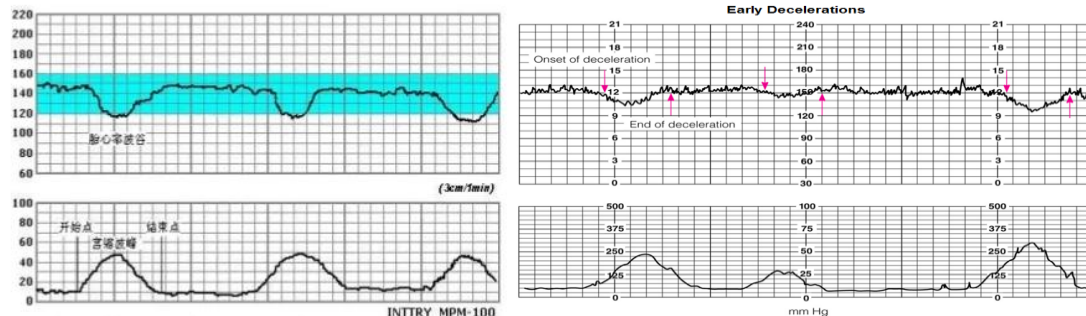
1. Ultrasound
2. Abdominal palpation
3. Uterine contraction
4. Music

Note: 120 – 160 = normal / more than 160 = tachycardia – Call Dr.

## Intervention

Document or continuing monitoring

## Early Deceleration. (Good finding)



Document or continuing monitoring

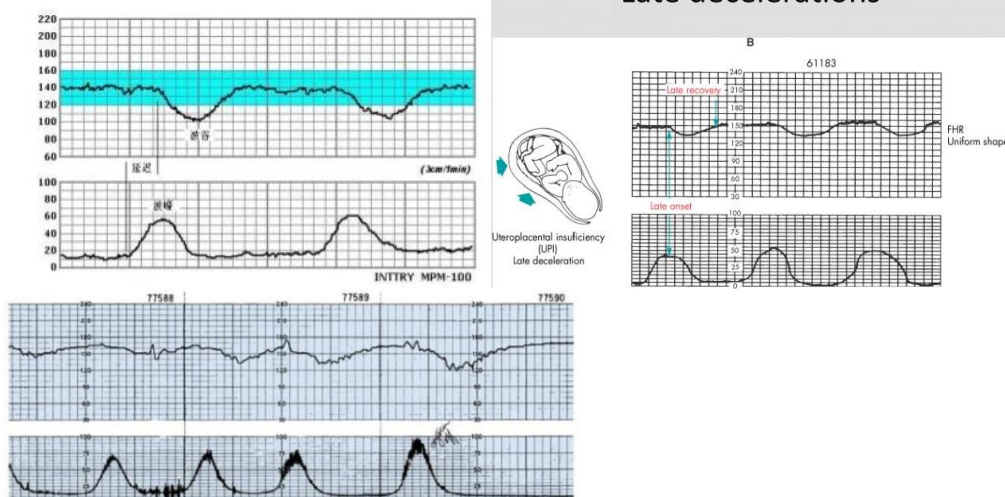
### Cause:

- Head compression

### Concept:

- Decreased FHR below baseline
- The rate at the lowest point of deceleration usually remains greater than 100 b/m and returns to baseline.
- Mirror image: cuando sube el pico de contracción de la mama, baja el FHR pero no mas de 100
- At the beginning of the contraction or at same time.

## Late Deceleration. (Priority=No good) (key = After)



Non-reassuring = Decreased baby O2 = uterus placental insufficiency = #1 Priority.

### Cause

- ✓ Pre-Eclampsia/Eclampsia

- ✓ Contraction stress test
- ✓ Maternal hypertension
- ✓ Maternal Diabetes
- ✓ Maternal Infection

Fetal FHR decelerated **AFTER** peak of uterine contraction.

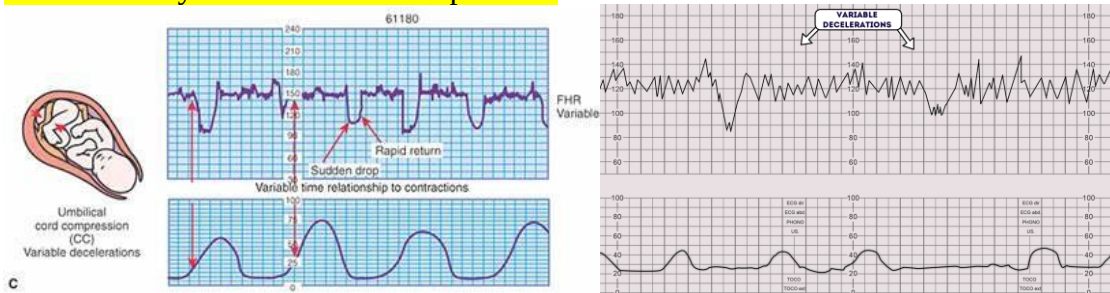
- Remain over 100 FHR
- Does not return to baseline

### **Intervention.** (board) **STOP**

1. Stop Oxytocin
2. Turn position to left or side lateral
3. O2 (8-10 L/min)
4. Physician or call Dr
- IV fluid

### **Variable Deceleration**

- Is cause by umbilical cord compression



### **Cause:**

- Umbilical cord compression
  - Short cord
  - Knot cord
  - Umbilical cord prolapses
  - Prolapse through vagina

### **Concept:**

- Decreased abruptly and deeply (fall & rise abruptly)
- Have V or W appears
- Occurs at time unrelated to contraction
- FHR decelerates less than 100b/min

### **Priority:**

- When FHR repeatedly decrease less than 70 b/min and persist at this level for at least 60 seconds = Sufrimiento Fetal = **Non reassuring**

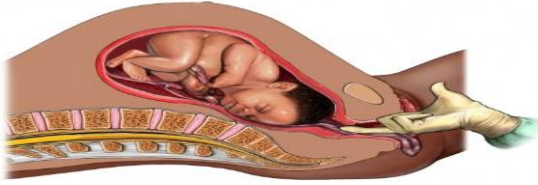
### **Intervention**

Umbilical Cord compression/umbilical cord prolapse and prolapse through the vagina

1. Stop Oxytocin
2. Change position:
  - ✓ Knee chest position
  - ✓ Trendelenburg
  - ✓ **Recumbent**
  - ✓ Side lateral
3. O2

4. Call the Dr.
5. IV fluids
- Amnioinfusion = Instill NSS to releave umbilical cord compression.

### Umbilical cord prolapsed



1. Put sterile gloves and introduce 2 fingers into the vagina and exert upward pressure (against to presentation part) while another nurse call for assistance.
2. Do not reintroduce the umbilical cord (if mention towel roll = OK)

### If prolapsed through vagina:

- a. Cover or wrap with moist sterile dressing.

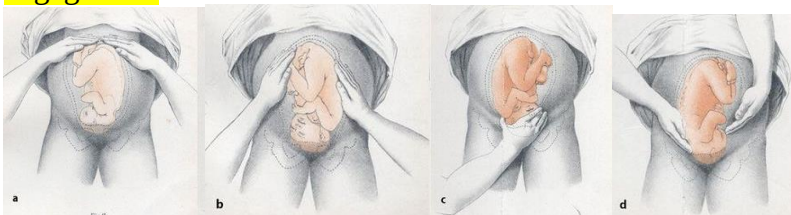
### Labor

Note: los dolores de parto se corren de la espalda para adelante.

| False Labor                                                                                                                                                                                                                                                                        | True labor                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ol style="list-style-type: none"> <li>1. Discomfort localized in abdomen or groin</li> <li>2. Contraction decreased intensity with ambulation.               <ul style="list-style-type: none"> <li>✓ Irregular without progress.</li> </ul> </li> <li>3. Cervix close</li> </ol> | <ol style="list-style-type: none"> <li>1. Contraction occurs regularly- increased with ambulation               <ul style="list-style-type: none"> <li>✓ Become stronger</li> <li>✓ Last longer</li> <li>✓ Closer together</li> </ul> </li> <li>2. Pain in lower back radiated to the abdomen “girdling” (around the waist)</li> <li>3. Progressive cervical dilation + Effacement (borramiento del cuello)</li> </ol> |

### Leopold’s maneuvers

Abdominal palpation = Use to determine Fetal presentation, lie position and engagement



1. **First Leopold maneuver (Fundus grip)**
  - Patient in supine position, placed both hands cupped over fundus and palpate to determinate:
    - ✓ Button’s (Brech) = Soft, immovable, large (nalgas up)
    - ✓ Vertex or occipitals = Hard, small, movable (cabeza up)

Board: en la primera maniobra se busca que hay en el fundus (cabeza o nalgas)

2. **Second Leopold maneuver (espalda – manos /pies)**
  - Placed one hand firmly on the side and palpate with other hand to determine the small parts and fetal back.



- ✓ Small parts: (knots or irregulars)
- ✓ Fetal back: (Smooth and hard)

Nota: HR se ausculta on the fetal back, en la 2 maniobra se busca espalda = smooth and hard

### 3. 3<sup>rd</sup> Leopold maneuver (to determine presentation part/ Fetal position)

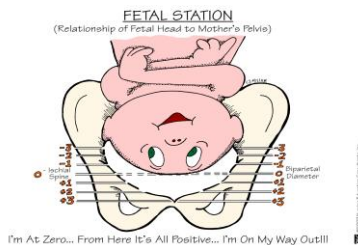
- Facing patient, grasp the area over symphysis pubis with the thumb and fingers and press to determine presentation.
  - Cephalic
  - Breech

### 4. 4<sup>th</sup> Leopold maneuver (engagement)

- Facing the client feet outline the fetal presentation part, with the palmar surface of both hands to determine degree of descent and attitude of the fetus

Nota: Este es la única maniobra que se hace por detrás del paciente

### Stations:

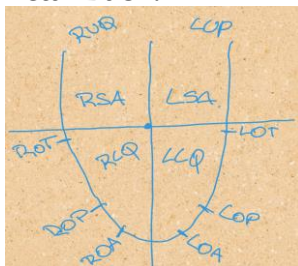


Ex:

Tiene 10 cm de dilatación, 90% effacement and -2 station

8cm dilation, 90 % effacement and +2 station = esta es la priority

### Fetal Back:



L delante = FHR in the Left side

R delante = FHR in the right side

O = en el medio cabeza ↓ and the FHR = below the umbilicus

S= en el medio = Sentado = FHR above the umbilicus.

Ex:

- RSA = FHR in right upper quadrant
- LSA = left upper quadrant
- ROA = right lower quadrant

EX: -First Leopold maneuver, nurse palpate soft, immobile and large mass = (nalgas up)

And In the second Leopold maneuver palpate a smooth hard area to the right

Dónde va el FHR = (RLQ)

Nota: Si tiene la boca para la derecha se ausculta el FHR para el lado contrario

## Stage of Labor

1. First stage = Start with the true labor  
Ends with the full dilation = 10cm and 100 % effacement =(transition)
2. Second stage = 10 cm of dilation, 100% effacement until baby born
3. 3rd stage = Start with baby born until placenta expulsion
4. 4rd stage = 4 h after placental expulsion
5. Post partum = hasta 30 dias after

### - Stage 1

- o Start with the true labor
- o Ends with the full dilation = 10cm and 100 % effacement =(transition)
- ✓ Latent Phase
  - o Start with true labor – until 0-3cm of dilation
  - o Can last 14 – 20 hours (amigos a ver el deporte)
  - o Uterine contraction every 15- 30 min of frequency
  - o 15 -30 seconds of duration
  - o Mild anxious, conversant.

Nota: contractions > 90 second = rupture de utero

Boar: husband watch football game in this stage.

### Intervention

1. Encourage mother and partner to participate in care.
2. Change position and ambulation
3. Offer fluids and Ice chips
4. Encourage voiding every 1 – 2 hours

Nota: una vez que se rompe la membrana ya no puede caminar y debe estar en lithotomy position.

Se le ofrece pain anesthesia = Mid and Active phase.

- ✓ Active phase
  - o Cervical dilation 4 – 7 cm
  - o Uterine contraction occur every 3-5 min
  - o 30 – 60 seconds of duration

### Intervention

1. Quiet enviroment
2. Promote confort
  - o Back rubs
  - o Sacral pressure
  - o Pillows support
  - o Position change
3. Instruct partner in effleurage (light stroking on abdomen) (massage en la barriga)
4. Offer fluids and ice chips
5. Encourage to voiding every two hours
6. Offer analgesic or anesthesia (mid- Active) 5-6 cm= Epidural block

- ✓ Transition phase
  - o Cervical dilation 8-10
  - o Uterine contraction 2-3 min
  - o 45 -90 second in duration = Strong intensity
  - o 100 % effacement

### Complication



- Respiratory Alkalosis
  - o Paresthesia: Numbness, tingling, burning sensation on the extremities and lips. (Lips and tip of fingers)
  - o Mouth stiffness = (boca apretada)
  - o Dizziness

#### **Intervention**

- ◆ Breathe in paper bag, hand cup or use partial Re-breather mask

- Respiratory techniques to prevent Respiratory alkalosis = Respiratory Pattern
  - o Pan – Pan – Blow
  - o Hee – Hee- Whow

#### **Intervention**

- Encourage rest between contraction
- Provide privacy
- Offer fluids and ice chips
- Encourage to voiding every 2 hours until membrane rupture.

**Board:** Cuando se pone a paciente en lithotomy? = at the end of 1st or beginning 2 do = when urge to push.

- **Second Stage of labor**
  - o 10 cm of dilation, 100% effacement until baby born
  - o Uterine contraction every 2 -3 min and lasting 60 – 75 seconds = Strong (se puede hacer caca aqui)

#### **Signs**

1. Urge to push or bearing down
2. Increased in bloody show
3. Bulging perineum and Anus

#### **Intervention**

1. Place client in Lithotomy position
2. Administer pudendal anesthetic (para cortar alla abajo)(Episiotomy)
3. Assessment every 5 min
4. Check for FHR
5. Assess for Amniotic fluid (if yellow or green) = Call the Dr. Bc risk for aspiration or pneumonia y hay que aspirar al bb

- **Third Stage of labor**
  - o Start with baby born until placenta expulsion, (no se jala)

**Signs** (que te indica que esta entrando al tercer stage)

1. Gush of blood
2. Mother describe a full feeling in vagina
3. Lengthening of umbilical cord

After placenta exposure:

Administer oxytocin after placenta expulsion to help uterus contraction (to prevent uterine Atonia) = el utero no se contrae.

- **Fourth Stage of labor**
    - o 1-4 hours after placental expulsión
- (3 and 4 stage)

1. **Vital sign**

➤ Increased temperature during 1st 12-24 hours = expected = 100.3 – 100.4 = Due to mild dehydration

- Administer IV fluid
- HR less than 60 = Expected
- Expective bradycardia (54-58)
- BP = if less than 90/60 mmhg = Call Dr

## 2. Check fundus

- ✓ During 12-24 h
  - At the umbilicus or below, at the midline and firm
- Boggy 1-2 cm above the umbilicus, and deviated to the right: = Full bladder
1. Encourage to void
  2. Massage
- Boggy 2cm above the umbilicus
- Massage.

## How to massage

- One hand remains cupped against uterus at the level of SP to support uterus and the other hand is upper to massage towards lower segment.



Board: Que esta mal? = RN did not support lower uterus (has to be supported to avoid uterus inversion.= complication.

## 3. Check for Hemorrhage

- More than 500 ml/24h for vaginal delivery
- More than 700/ 24 h for C- Section

**LOCHIA.** (la rubia esta celosa de Alba)

### Rubra

- Bright red with clots, fleshy odor (0-3 days)
  - Follow up if more than 3 days

If 5 days and have rubra = call Dr

### Serosa

- Brownish pink with fleshy odor (4-10 days)

### Alba

- Yellow to white 11 – 14 days

If pad saturate in less than 1 hour = Hemorrhage = Call Dr

Check the last time que se lo cambiaron.

## 4. Dry and suction infant

- ✓ Perform APGAR score

5. Place blanked on mother abdomen and allow skin to skin contact

**Post-partum** (After 4h from birth)

Gently cleanse the vulva with surgigator or peri bottle (front to back.)

Note: (no tocar el area vaginal with the surgigator bc infection)

### **Post-partum Discomfort**

1. After birth pain = Expected due to uterine contraction, = analgesic before breast feeding.

Stimulate by:

- a. Breast feeding
- b. Oxytocin

2. **Perineal Discomfort**

- Ecchymosis, swelling:
  - o Ice pack during first 24 hours
  - o After 24 hours = sitz bath for 20 min 3 -4 times/day

Nota: In case of laceration = Avoid Enemas and suppositories

### **Breast**

- Breast feeding Position
  - o Football position or belly to belly
- The sucking sequency and proper latch- on
  - o Nipple and areola are down into mouth enough for:
  - o Lip to cover 1- 1.5 inch of areola
  - o Jaw should move up and down in a rhythm motion during milk transfer
  - o Ears wiggle
- Cheeks should be full and rounded (no sucked in).
- The best indicator of proper milk transfer is audible swallowing “K-ah” sound = Board
- Release infant mouth from nipple by inserting one finger to break suction.
- Start for the last one

Nota: Se alimenta cada vez que quiera con un máximo de 3 hours.

### **Engorgement**

| Non- Nursing woman                                                                                                                                                                                                                                                                            | Nursing women                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>- HIV</li><li>- Galactosemia</li><li>- Alcohol or drug abuse at the moment.</li><li>✓ Avoid nipple stimulation</li><li>✓ Apply breast binder and wear snug- fitting bra</li><li>✓ Apply ice pack and analgesic</li><li>✓ Resolve 24- 36 hours</li></ul> | <ul style="list-style-type: none"><li>- Feeding Frequently</li><li>- Ice pack or Ice cabbage leaves between feedings. (hoja de col congelada= ok)</li><li>- Warm shower or warm compress before breast feeding</li><li>- Analgesic 30 min before.</li></ul> |

### **Plugged Duct**

- Heat and massage (bc conductos tupidos)

### **Nipple irritation.** (board)

- Clean with water (NO soap) and dry.
- Apply breast milk to the nipple and areola after each feeding and air dry.

### **Expression of milk**

- Out of refrigerator – 4 hours

- Refrigerated = 2-3 days
  - Freezer = 2 weeks
  - Deep Freezer = 2- 6 months
- (Plastic pomito donde la almacenas: Board)

### **Psychosocial Adjustment**

- Bounding signs
  - o Touching baby face with fingertips. (board)
  - o Naming
  - o Direct eye contact
  - o Kissing
  - o Smiling

### **- Phases of Adjustment**

| Taking In                                                                                                                                                                                                                                                                                                                                          | Taking Hold                                                                                                                                                                                                                                                       | Letting Go                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>- 1-2 days after birth<br/>(se centra solo en sus problemas)</li> <li>- Dependency</li> <li>- Only preoccupied with self-own needs<br/>(food and sleep)</li> <li>- Talkative</li> <li>- Hesitate to make decision</li> <li>- Re-tells birth perception</li> <li>- Ask for help in a simple task.</li> </ul> | <ul style="list-style-type: none"> <li>- 2-10 days after birth<br/>(acepta sugerencia de la suegra = board)</li> <li>- Dependency – Independence</li> <li>- Open to instruction</li> <li>- Performing self-care and express concern for self and baby.</li> </ul> | <ul style="list-style-type: none"> <li>- 5-6 weeks after birth</li> <li>- Independence</li> <li>- Assume a new roll responsibility.</li> <li>- Adjustment and accommodation infant in family</li> <li>- Ex: Arma la cuna.</li> </ul> |

### **1. Post- partum complication**

| Post-partum blue<br>3-7 days postpartum                                                                                                                                                                                                                            | Post- partum depression<br>More than 2 weeks                                                                                                                                                                                                                                                                                                                                                                                      | Post- partum Psychosis<br>At any time                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>- Normal occurrence of roller coaster sensation of emotion.</li> <li>- Weeping</li> <li>- Insomnia</li> <li>- Emotionally labile</li> <li>- Fatigue</li> </ul> <p><b><u>Intervention</u></b></p> <p>1 Emotional support</p> | <ol style="list-style-type: none"> <li>1. Anhedonia = loss of joy</li> <li>2. Less responsive to the infant.</li> <li>3. Asocialization</li> <li>4. Apathia</li> <li>5. Frequent crying and sadness</li> <li>6. Lack about physical appearance.</li> <li>7. Suicidal ideation</li> </ol> <p><b><u>Treatment</u></b></p> <p>1. SSRI – except for :<br/>Fluoxetine = Contraindicated in Breast feeding.<br/>Paroxetine = choice</p> | <ul style="list-style-type: none"> <li>- Emergency = hospital admission.</li> <li>- Personal history of psychotic disease</li> </ul> <p><b><u>Assessment</u></b></p> <ol style="list-style-type: none"> <li>1. Hallucination</li> <li>2. Illusion</li> <li>3. Delusion</li> <li>4. Break with reality</li> <li>5. Panic</li> </ol> |

### **2. Hematoma**

- Collection of bleeding in the tissue.

### **Cause**

- Trauma- forceps or injury of blood vessels
- Bulbar hematoma (board)
  - o Pain in the vulva

### **Assessment**

1. Abnormal severe vulvar or vaginal pain
2. Pressure in the perineal area = Report bc hypovolemic shock
3. Sensitive, bulging mass in the perineal area with skin discoloration.
4. Inability to void.
5. Decreased HB or Hematocrit
6. Signs of shock in case of rupture.

#### **Intervention**

1. Apply ice pack
  2. Analgesic
  3. Blood transfusion set at the bed side.
  4. Antibiotic prophylactic
  5. Prepare for incision and drainage.
- 
3. **Endometritis.** (Standard precaution)
    - Inflammation and infection of Endometrium

#### **Assessment**

1. Fever + chills
2. Rubra persistent > 3 days with foul smelling.
3. Pelvic discomfort
4. Anorexia
5. Uterine slowly subinvolution

**Note:** Normal = decrease 1 cm/day, 10-12 days after birth at the SP level

#### **Intervention. (hospital admission)**

1. Plenty fluid
2. Antibiotic after culture
3. Fluid

#### **4. Mastitis**

- Inflammation of the breast due to Staphylococcus infection

#### **Assessment**

1. Fever + chills
2. Localized heat and swelling
3. Pain, tender axillary lymphonodes
4. Flu like symptoms (fatigue, malaise)

#### **Treatment**

1. Standard precaution, frequently hand washing.
2. Apply ice pack or cold compress
3. Maintain lactation or encourage manual expression (board)  
**Continue lactando en el lado de la mastitis.**
4. Antibiotic
5. Analgesic

#### **5. Shoulder distocia**

- After vaginal delivery of the baby head, the anterior shoulder get caught above the mother pubic bone = “Turtle Sign”

#### **Intervention. (Shoulder dystocia)**

1. Mc Robert Maneuver
  - Flexing and abducted the maternal hips, positioning the maternal thigh up onto maternal abdomen.
2. Suprapubic Pressure

6. **Rupture of the uterus.** (life threatening)

**Cause**

- Uterine contractions more than 90 seconds
- Oxytocin
- Trauma
- C- Section

**Assessment**

1. Hypovolemic shock (hypotension + tachycardia)
2. **Contraction sudden stops**
3. Absent FHR
4. Board like abdomen radiated to the shoulder

**Intervention**

1. Surgery = C-Section

7. **Uterine Inversion.** No. 1 Priority

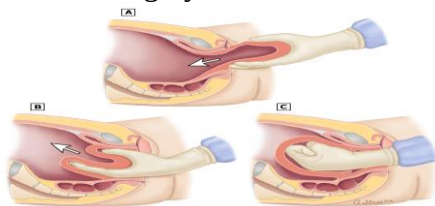
- Uterus turn inside out

**Assessment**

1. **Depression in the fundal area**
2. Anterior of the uterus may be seen through cervix or protruding through vagina.
3. Severe pain
4. Hypovolemic shock

**Intervention**

1. Manual reintroduction
2. Surgery



**Diabetes Gestational**

- **Screening 24 – 28 weeks**
- Gold standard test
  - 1 hour glucose challenge test (no ayunas) = if positive then do:
  - Glucose tolerance test. = confirmatory and mas profundo

**Intervention** (diabetes gestational)

1. Diet and exercise
  2. Regular insulin (RRR) (IV)
- Nota; No Oral hypoglycemiante, (son teratogenic)

- Risk for:
    - Hypoglycemia maternal
    - Macrosomal fetus weight > 400g
- Nota: el bebe también tiene riesgo for hypoglicemia

- 1st Trimester = Decreased Insulin requirement
- 2<sup>nd</sup> Trimester = Begin to Increase
- 3<sup>rd</sup> Trimester = 3- 4 x initial dose

- Day of birth = decreased drastically